

## Homework (Habanero)

- Q1.** Simplify  $\sqrt{12} + \sqrt{3}$   
(A)  $3\sqrt{3} + \sqrt{3}$   
(B)  $4\sqrt{3}$   
(C)  $\sqrt{15}$   
(D)  $2\sqrt{3}$   
(E)  $\sqrt{12} + 1$
- Q2.** A patient is prescribed  $2\sqrt{5}$  milligrams of medication. If the patient already took  $\sqrt{5}$  milligrams, how much more medication does the patient need?  
(A)  $\sqrt{5}$  milligrams  
(B)  $2\sqrt{5}$  milligrams  
(C)  $3\sqrt{5}$  milligrams  
(D)  $\sqrt{10}$  milligrams  
(E) 5 milligrams
- Q3.** Simplify  $\sqrt{24} \cdot \sqrt{6}$   
(A)  $12\sqrt{2}$   
(B) 12  
(C)  $\sqrt{144}$   
(D)  $6\sqrt{2}$   
(E)  $\sqrt{36}$
- Q4.** The number of cells in a culture grows at a rate of  $2\sqrt{t}$  per hour, where  $t$  is the time in hours. If the initial number of cells is  $\sqrt{2}$ , how many cells will be present after 2 hours?  
(A)  $4 + \sqrt{2}$   
(B)  $2 + \sqrt{2}$   
(C) 6  
(D)  $4\sqrt{2}$   
(E) 8
- Q5.** Simplify  $\frac{\sqrt{18}}{\sqrt{2}}$
- (A)  $\sqrt{9}$   
(B)  $3\sqrt{2}$   
(C)  $\sqrt{18}$   
(D)  $\frac{\sqrt{18}}{2}$   
(E)  $\sqrt{8}$
- Q6.** A medical researcher is analyzing data on the effectiveness of a new treatment. The data shows a correlation of 0.8 between the treatment and patient outcomes, measured in units of  $\sqrt{5}$ . If the researcher wants to calculate the correlation in terms of  $\sqrt{2}$ , what would be the equivalent value?  
(A) 1.6  
(B)  $0.4\sqrt{10}$   
(C)  $1.1\sqrt{2}$   
(D)  $0.8\sqrt{2}$   
(E) 1.3
- Q7.** Simplify  $(2\sqrt{3} + 1)(\sqrt{3} - 1)$   
(A)  $5 - 2\sqrt{3}$   
(B)  $5 + 2\sqrt{3}$   
(C) 1  
(D) 7  
(E)  $2\sqrt{3} - 1$
- Q8.** The probability of a certain medical condition occurring is  $\frac{1}{\sqrt{2}}$ . If the probability of the condition occurring twice is  $\frac{1}{2}$ , what is the probability that it will occur at least once?  
(A)  $\frac{1}{\sqrt{2}}$   
(B)  $\frac{1}{2}$   
(C)  $\frac{3}{4}$   
(D)  $\frac{1}{4}$   
(E)  $\frac{2}{3}$

### Open Ended Questions (FRQ)

- Q9.** In a study on the effects of a new medication, the average response time was  $3\sqrt{2}$  minutes. If the standard deviation of the response times was  $\sqrt{5}$  minutes, what percentage of patients responded within one standard deviation of the average?
- Q10.** A pharmaceutical company is developing a new drug that has a  $2\sqrt{3}$

### Chef's Tips

- Tip 1: Make sure to simplify all radical expressions before solving the problem
- Tip 2: Pay attention to units when working with word problems
- Tip 3: Double-check calculations to ensure accuracy